

Plan: GömbSoma — compositional hypergraph hierarchy for the sensorimotor stack

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1 Framing: GömbSoma is additive to Gömb, not a replacement

Gömb (Hungarian: *sphere*; three orthogonal shells: OuterFIRShell, MiddleHSiKAN, InnerCPML-Core) achieves Epinions 0.9526 ± 0.0018 strict protocol and 0.9527 ± 0.0018 unrestricted — the architecture saturates its structural signal on social-trust graphs. We hypothesise Gömb is the *cognitive-level* architecture: it operates on abstractions (vertices are agents, edges are trust relationships) where the three axes (geometric, learnable nonlinearity, routing) factor naturally.

GömbSoma (Greek *soma*: body; this plan) is a different layout of the same primitives, targeted at the *sensorimotor level*: where the input is raw or near-raw and structure must be built up compositionally from short open walks $\rightarrow$ closed polygons $\rightarrow$ balanced triangles $\rightarrow$ derivative-nodelet abstractions. Gömb stays canonical; GömbSoma ships as a parallel branch in `signedkan_wip/src/hymeko_gomb/soma/` and is empirically falsifiable against Gömb on a curated test battery.

The naming reflects a deliberate division of labour: Gömb (the sphere; abstract / cognitive) sits above GömbSoma (the body; sensory *and* motor). A mature system would compose the two — GömbSoma processes raw sensory input and emits motor commands, Gömb reasons about agent-level abstractions on top.

2 Scope and goal

Build and test the hypothesis that a compositional hierarchical hypergraph network — walk-layer $\rightarrow$ polygon-layer $\rightarrow$ triangle-layer, with Clifford-FIR as the inter-layer transfer operator — outperforms the Gömb flat-mixture shell on tasks where the input has spatial / compositional / sequential structure not yet abstracted (vision, kinematic graphs, formal language detection), while matching Gömb on already-abstracted signed-link benchmarks.

3 Architecture

3.1 Layer types

Layer	Primitive	Closure / property
Walk-conv	open walk w_k , $k = 2, 3, \dots$	none (open)
Polygon-conv	closed cycle c_k , $k \geq 4$	one ring-closure
Triangle-conv	c_3 with Cartwright–Harary gate	full structural balance check
Abstraction-conv	derivative-nodelet contraction	balanced-clique super-vertex

3.2 Inter-layer Clifford-FIR transfer

A single Clifford-FIR module per inter-layer boundary, parametrised by the grade transition: walks live in grade-1 (vector-shaped), polygons in grade-2 (bivector-shaped), triangles in grade-3 (trivector-shaped). The Clifford geometric product implements the grade-lifting operator that promotes one layer's feature into the next layer's natural representation.

3.3 Compositional rule

Each closure transition is strictly stronger than the previous:

$$\text{walk} \xrightarrow{+\text{ring closure}} \text{polygon} \xrightarrow{+\text{Cartwright-Harary}} \text{balanced triangle} \xrightarrow{+\text{contraction}} \text{super-vertex}.$$

Walks subsume nothing closed; polygons subsume open walks (as cycle without one edge) plus a closure constraint; balanced triangles subsume polygons plus a balance constraint; super-vertices subsume balanced cliques plus a contraction.

4 Affected files

All under `signedkan_wip/`, outside any `CORE.YAML` lockdown:

- `src/hymeko_gomb/soma/__init__.py` — package boundary.
- `src/hymeko_gomb/soma/hg_conv.py` — HypergraphConv ABC + canonical implementation; the message-passing primitive shared by all layer types.
- `src/hymeko_gomb/soma/walk_layer.py` — WalkConvLayer (HG-conv over $\{w_k\}$).
- `src/hymeko_gomb/soma/polygon_layer.py` — PolygonConvLayer (HG-conv over $\{c_k\}_{k \geq 4}$).
- `src/hymeko_gomb/soma/triangle_layer.py` — TriangleConvLayer (HG-conv over c_3 + Cartwright-Harary balance gate).
- `src/hymeko_gomb/soma/inter_layer_clifford.py` — InterLayerCliffordFIR (grade-lifting Clifford-FIR; replaces the Gömb OuterFIRShell role).
- `src/hymeko_gomb/soma/abstractions.py` — DerivativeNodeletContraction (balanced-clique $\rightarrow$ super-vertex contraction).
- `src/hymeko_gomb/soma/hierarchy.py` — GombSomaHierarchy (top-level cascade).
- `src/run_gomb_soma_smoke.py` — entry point, kwargs-compatible with `run_gomb_smoke.py` so the same orchestrator scripts apply.
- `tests/test_gomb_soma_walk_layer.py`
- `tests/test_gomb_soma_polygon_layer.py`
- `tests/test_gomb_soma_triangle_layer.py`
- `tests/test_gomb_soma_hierarchy.py`
- `tests/test_gomb_soma_inter_layer_clifford.py`

No edits to Gömb files. No edits to `run_gomb_smoke.py`.

5 CORE.YAML items touched

None. All GömbSoma code lives in `signedkan_wip/`.

6 Interface changes

Public: a new entry point `run_gomb_soma_smoke`, sharing all CLI flag names with `run_gomb_smoke` that make sense (`-dataset`, `-seed`, `-n-epochs`, `-device`, `-edge-split`, `-lr`, `-unrestricted-cycles`, `-shuffle-train-signs`). Removed: shell-specific flags (`-M-outer`, `-d-outer`, `-n-tiers`, ...). Added: `-n-walk-layers`, `-n-polygon-layers`, `-n-triangle-layers`, `-max-walk-len`, `-max-polygon-len`, `-abstraction-depth`.

No changes to Gömb’s public interface or wire protocol.

7 Test strategy

7.1 Unit (per layer)

For each of WalkConv / PolygonConv / TriangleConv:

1. permutation equivariance: `layer.forward(perm(H)) == perm(layer.forward(H))` up to numerical tolerance.
2. sparse-aware: layer doesn’t materialise dense incidence matrices.
3. signed-correct: layer respects σ -product polarity (positive cycles route to one sub-bank, negative to another).
4. parameter count: $< 100\text{ k}$ per layer at default hidden width 16.
5. gradient flow: `loss.backward()` produces non-zero gradients on every parameter (no dead branches).

7.2 Unit (inter-layer Clifford-FIR)

1. grade transition correctness: input grade-1 multivector $\rightarrow$ output grade-2 multivector; verify the Clifford geometric product structure.
2. rotation equivariance on a synthetic 3D-embedded graph: rotate the input, check the output rotates accordingly within the grade.

7.3 Integration (hierarchy)

Synthetic SBM with known cycle structure: 4-community graph at $n = 200$ with planted triangles. GömbSoma hierarchy must:

1. recover community assignment at ≥ 0.85 ARI in ≤ 60 epochs.
2. show walk-layer activates on cross-community edges; polygon-layer activates on within-community closures; triangle-layer activates on planted triangles. Verified via per-layer activation logging.

7.4 Falsification battery

Four datasets, Gömb vs GömbSoma paired, 5 seeds each:

Dataset	Gömb baseline	GömbSoma prediction
Bitcoin Alpha	0.8972 ± 0.0079	match (cognitive task)
Slashdot	0.9017 ± 0.0008	match (cognitive task)
Epinions strict	0.9526 ± 0.0018	match (cognitive task)
Kinematic 13-URDF	1.000 family acc, $k = 4/k = 6$	match or improve
HymeYOLO +ricci+kcycle	$\sim 0.20 \pm 0.04$ mAP50	improve (sensory task)
DFA-augmented chain (synthetic)	no Gömb baseline	> 0.95 accuracy

8 Performance budget

Each GömbSoma layer is a single hypergraph convolution: linear in $|E| + \sum_k |C_k|$, comparable per-layer cost to one Gömb shell. With 4 layers (walk, polygon, triangle, abstraction) the total compute is $\sim 4/3 \times$ Gömb.

- Peak RSS: ≤ 12 GB (cgroups MemoryMax 16 GB).
- Per-seed wall: ≤ 10 min on Epinions ($v_1 = 6.5$ min).
- GPU memory: ≤ 7 GB on the RTX 2070 SUPER.
- Total falsification battery (4 datasets $\times$ 5 seeds + 1 vision $\times$ 5 seeds + 1 synthetic $\times$ 5 seeds): ≤ 5 h.

9 Rollback path

GömbSoma is fully additive. To disable: delete `signedkan_wip/src/hymeko_gomb/soma/` and `run_gomb_soma_smoke.py`. Gömb entry points unchanged.

10 Risk anticipation

- *Performance-contract preservation.* Gömb already enforces `max_cycles`, top- k caps, ABB pruning. GömbSoma layers must inherit the same caps; we ship a single `EnumerationConfig` dataclass shared with Gömb’s enumeration entry points to prevent drift.
- *Worst-case input.* Epinions (842k edges, 132k nodes). The triangle layer’s Cartwright–Harary balance gate runs an additional $O(|C_3|)$ pass; we cap $|C_3|$ at the same `topk` per-vertex cap that Gömb uses.
- *Empirical falsification.* GömbSoma may match Gömb everywhere $\rightarrow$ redundant variant. We ship-deprecate GömbSoma if the 4-dataset paired delta is ≤ 0 everywhere. GömbSoma may underperform Gömb on sensory tasks $\rightarrow$ the compositional-hierarchy hypothesis is wrong; the plan dir is preserved as a negative-result anchor.
- *Empty-plan-dir hygiene.* If we shelve GömbSoma, the plan dir is deleted before commit; no half-finished implementations left behind.

11 Decision tree

- GömbSoma beats Gömb on sensory (vision, kinematic, sequence) but *matches* on signed-link $\Rightarrow$ confirms the sensory / cognitive layered hypothesis. Both stay shipping; documentation positions them as layered.
- GömbSoma beats Gömb *everywhere* $\Rightarrow$ Gömb is a flat approximation of GömbSoma; promote GömbSoma to default, Gömb stays as ablation reference.
- GömbSoma matches Gömb *everywhere* $\Rightarrow$ redundant variant; GömbSoma is documented as a negative result.
- GömbSoma loses everywhere $\Rightarrow$ compositional-hierarchy hypothesis is wrong; GömbSoma stays as a negative-result archive.

12 Phases (one per session, per CLAUDE.md §6.5 anti-pattern #11)

1. HypergraphConv ABC + unit tests. No layers yet.
2. Walk layer + unit tests + synthetic SBM smoke.
3. Polygon layer + unit tests + synthetic SBM smoke.
4. Triangle layer + Cartwright–Harary gate + unit tests + synthetic SBM smoke.
5. Inter-layer Clifford-FIR transfer + unit tests + grade-transition correctness test.
6. Top-level hierarchy + `run_gomb_soma_smoke.py` entry point + Bitcoin Alpha 1-seed production-scale smoke.
7. Falsification battery (4 datasets $\times$ 5 seeds + vision + synthetic) + report.

Each phase produces a working module under unit-test coverage. No phase combines.

13 Bottom line

GömbSoma is the next architectural experiment, not a replacement. We keep Gömb as the canonical signed-link benchmark architecture; we test GömbSoma specifically against the hypothesis that compositional walk $\rightarrow$ polygon $\rightarrow$ triangle $\rightarrow$ abstraction is the right sensory-level structure-building stack. The falsification battery is concrete and the decision tree is pre-registered.